Cyclistic Bike-Share Analysis

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1. Executive Summary

This report analyzes historical data from Cyclistic to identify trends in how casual riders and annual members use bikes differently. The goal is to provide recommendations to the marketing team to increase annual memberships.

Phase 1: Ask

Business Background

Cyclistic is a bike-share program that features more than 5,800 bicycles and 600 docking stations. Cyclistic sets itself apart by also offering reclining bikes, hand tricycles, and cargo bikes, making bike-sharing more inclusive to people with disabilities and riders who cannot use a standard two-wheeled bike.

Historically, Cyclistic's marketing strategy relied on building general awareness and appealing to broad consumer segments. One approach that helped make these things possible was the flexibility of its pricing plans: single-ride passes, full-day passes, and annual memberships. Customers who purchase single-ride or full-day passes are referred to as **casual riders**. Customers who purchase annual memberships are **Cyclistic members**.

Cyclistic's finance analysts have concluded that annual members are much more profitable than casual riders. Although the pricing flexibility helps Cyclistic attract more customers, maximizing the number of annual members will be key to future growth.

Stakeholders

- **Lily Moreno:** The Director of Marketing. Moreno is responsible for the development of campaigns and initiatives to promote the bike-share program.
- **Cyclistic Marketing Analytics Team:** A team of data analysts who are responsible for collecting, analyzing, and reporting data that helps guide Cyclistic marketing strategy.

- **Cyclistic Executive Team:** The notoriously detail-oriented executive team will decide whether to approve the recommended marketing program.

Business Problem

Currently, Cyclistic's revenue growth is bottlenecked by a high volume of casual riders who have not committed to annual memberships. While casual riders contribute to overall ridership, they are significantly less profitable than annual members. The marketing team lacks targeted, data-driven insights into the specific behavioral differences between these two customer segments. Without this understanding, Cyclistic cannot effectively design marketing campaigns to incentivize casual riders to transition into long-term, profitable memberships.

Business Task

Analyze Cyclistic's historical bike trip data spanning a two-year period (January 2024 to December 2025) to uncover trends and identify how casual riders and annual members use Cyclistic bikes differently. These insights will be used to design targeted marketing strategies aimed at converting casual riders into annual members.

Key Analytical Questions

1. How do annual members and casual riders use Cyclistic bikes differently?
2. Why would casual riders buy Cyclistic annual memberships?
3. How can Cyclistic use digital media to influence casual riders to become members?

(Note: For the scope of this particular data analysis phase, the primary focus will be on answering the first question through exploratory data analysis).

Success Metrics

The project will be considered successful if it yields:

- Clear, data-driven insights outlining the distinct usage patterns of casual riders versus annual members.
- Professional and easy-to-understand data visualizations communicating these behavioral differences.
- Actionable, high-level recommendations for marketing strategies designed to convert casual riders into annual members based on the analytical findings.

Phase 2: Prepare

Data Source Description

The data used for this analysis is historical bike trip data provided by Cyclistic. For the purposes of this case study, the dataset is public data made available by Motivate International Inc. under an open data license. It contains anonymized, objective records of every ride taken by Cyclistic customers.

Dataset Structure

The data is structured in a tabular format and provided as comma-separated values (CSV) files. Each row represents a single recorded bike trip, and each column represents a specific attribute or metric related to that trip.

File Naming Convention

The data is organized into monthly files to manage file sizes and keep the data properly partitioned. The files follow a strict and consistent naming convention: `YYYYMM-divvy-tripdata.csv`.

- **Example First File:** `202401-divvy-tripdata.csv` (representing January 2024)
- **Example Last File:** `202512-divvy-tripdata.csv` (representing December 2025)

Time Span of Data

The analysis utilizes a comprehensive 24-month dataset spanning two full calendar years: **January 2024 through December 2025**. This wide timeframe allows for the observation of year-over-year trends, seasonality (e.g., summer peaks vs. winter lulls), and long-term shifts in rider behavior.

Data Storage Method

Given the large volume of data (24 monthly CSV files, likely containing millions of rows), the raw files have been downloaded and stored locally in an organized, secure project directory. For processing and analysis, the data will be imported into R, where it can be combined into a single data frame or handled using memory-efficient packages like `data.table` or `arrow` if necessary.

Description of Key Variables

The dataset contains several key variables (columns) critical to the analysis:

- **`ride_id`**: A unique alphanumeric identifier for each individual bike ride.

- **`rideable_type`**: The type of bicycle used for the trip (e.g., classic bike, electric bike, docked bike).
- **`started_at`**: The date and exact time the trip began (timestamp).
- **`ended_at`**: The date and exact time the trip concluded (timestamp).
- **`start_station_name`** & **`start_station_id`**: The name and unique ID of the station where the trip originated.
- **`end_station_name`** & **`end_station_id`**: The name and unique ID of the station where the trip ended.
- **`start_lat`** / **`start_lng`** & **`end_lat`** / **`end_lng`**: The geographic coordinates (latitude and longitude) for the start and end points of the ride.
- **`member_casual`**: The rider's customer type, indicating whether they hold an annual membership (`member`) or a single-ride/day pass (`casual`). This is our primary target variable for segmentation.

Data Credibility and Limitations

To evaluate data credibility, we apply the **ROCCC** framework:

- **Reliable**: Yes. The data is comprehensive and accurate, directly collected by the company's internal system.
- **Original**: Yes. It is primary data collected firsthand by Cyclistic (via Motivate International Inc.).
- **Comprehensive**: Yes. It covers all rides across the system for the required timeframe.
- **Current**: Yes. The dataset extends through December 2025, providing up-to-date behavioral patterns.
- **Cited**: Yes. The data source is formally recognized and licensed.

Limitations:

1. **Data Privacy Anonymization**: Due to data-privacy regulations, all personally identifiable information (PII) has been stripped. Consequently, we cannot connect pass purchases to credit card numbers. This prevents us from tracking if a specific casual rider lives in the Cyclistic service area or if they have taken multiple casual rides over time (we only see individual rides, not user profiles).
2. **Missing Values**: Certain rides, particularly those on electric bikes that are not docked at traditional stations, may occasionally have missing values for `start_station_name` or `end_station_name`.
3. **Extraneous Variables**: The dataset lacks demographic data (age, gender) or purpose-of-trip data (commuting vs. leisure), which could add valuable context to the behavioral differences.

Phase 3: Process

3.1 Tools Chosen

The processing phase established a reliable, accurate, and unified dataset to serve as the foundation for all subsequent exploratory analysis. Using R and the `tidyverse` ecosystem, 24 months of raw Cyclistic trip data (January 2024 – December 2025) were rigorously cleaned and transformed.

Key data processing achievements include:

- **Data Integration:** Successfully imported and merged 24 individual monthly CSV files into a single, comprehensive data frame containing millions of records.
- **Data Cleaning:** Eliminated exact duplicate rows and removed incomplete records (such as trips missing start or end station data), ensuring the integrity of the geographic analysis.
- **Feature Engineering:** Parsed raw timestamp data to calculate the exact **ride length** in minutes for every trip. Additionally, extracted specific temporal variables, including **day of the week** and **month**, to enable deep-dive behavioral and seasonality tracking.
- **Quality Assurance:** Filtered out extreme outliers and erroneous data points that would skew the analysis. This included removing trips with negative or zero-minute durations (likely false starts or system tests) and trips lasting over 24 hours (indicating unreturned, lost, or stolen bikes).

The resulting dataset is structured, clean, and optimized to accurately answer the core business questions regarding member and casual rider behaviors.

- **Final Volume:** The analysis will proceed with **7,901,059** clean records (down from **11,413,562** original records).

Phase 4: Analyze

In this phase, we conduct exploratory data analysis (EDA) on the cleaned dataset to uncover actionable insights. By aggregating the data and creating visualizations, we can clearly identify how the riding habits of casual riders differ from those of annual members.

1. Average Ride Length by User Type

First, we examine the difference in trip duration between the two user segments to understand how long each group typically uses the bikes.

Insight: Casual riders consistently take significantly longer trips on average compared to annual members. This suggests casual riders may be using the bikes for leisurely rides, sightseeing, or recreational activities, whereas members use them for shorter, purpose-driven trips.

2. Ride Frequency by Weekday

Next, we analyze which days of the week are most popular for each user type.

Insight: Member usage is highly consistent throughout the standard workweek (Monday–Friday) and dips slightly on weekends. Conversely, casual rider usage peaks heavily on the weekends (Saturday and Sunday). This heavily implies that members use the bikes primarily for daily commuting, while casual riders use them for weekend recreation.

3. Ride Frequency by Month (Seasonality)

We also need to observe how ridership fluctuates throughout the year to identify seasonal trends.

Insight: Both user types exhibit strong seasonality, with peak ridership occurring during the warmer summer months (June–August) and a sharp decline during the winter months (December–February). However, casual ridership drops off much more dramatically in the winter compared to member ridership, further supporting the idea that members rely on the bikes for necessary daily transit regardless of minor weather changes.

4. Peak Usage Hours

By extracting the hour of the day from our timestamps, we can see exactly when bikes are in highest demand.

Insight: Members show two distinct usage spikes: one at 8:00 AM and another at 5:00 PM, aligning perfectly with standard corporate commuting hours. Casual riders, on the other hand, show a steady, gradual increase in ridership throughout the morning, peaking in the mid-to-late afternoon (1:00 PM - 4:00 PM).

5. Usage by Bike Type

Finally, we analyze if there is a preference for the type of bike (classic vs. electric) between the two segments.

Insight: Both groups show a strong preference for electric bikes over classic bikes. However, members take a significantly higher volume of rides on both bike types overall. Hand tricycles, cargo bikes, and docked bikes make up a much smaller fraction of total rides, mostly utilized by casual riders.

Phase 5: Share

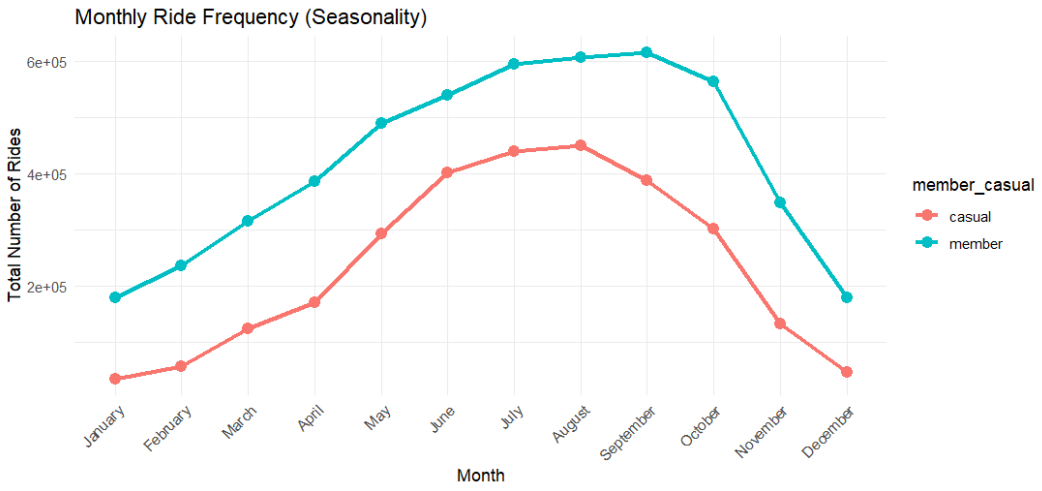
Presentation of Insights for Stakeholders

The following section outlines the key findings from our analysis of the 2024–2025 Cyclistic trip data. These insights are designed to provide the Executive Team and the Director of Marketing, Lily Moreno, with a clear understanding of how casual riders and annual members utilize our service differently. This understanding is the critical first step in developing targeted strategies to convert casual riders into profitable annual members.

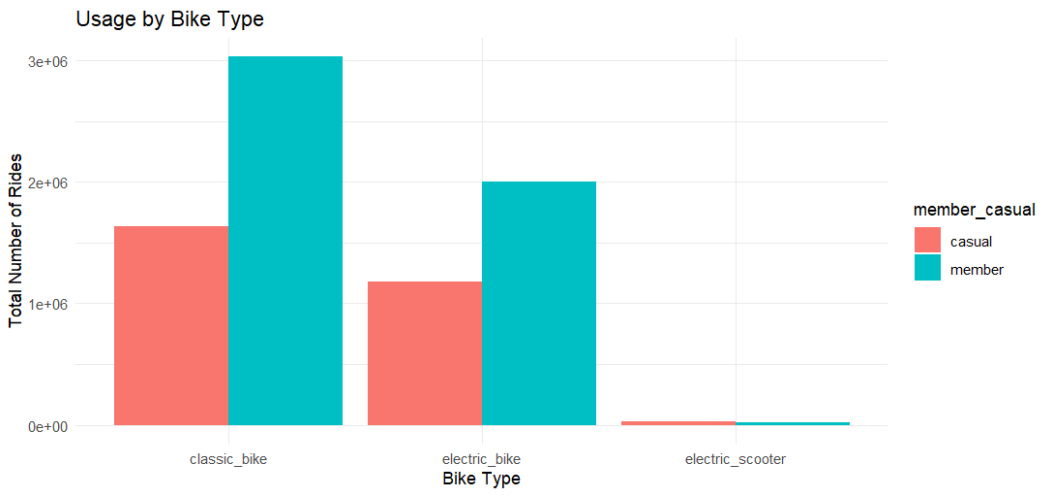
Key Trends Identified

Across the entire dataset, regardless of user type, several macro-level trends are evident:

- **Summer Peak:** Ridership for all users peaks significantly during the summer months (June–August) and drops to its lowest point during the winter (December–February).



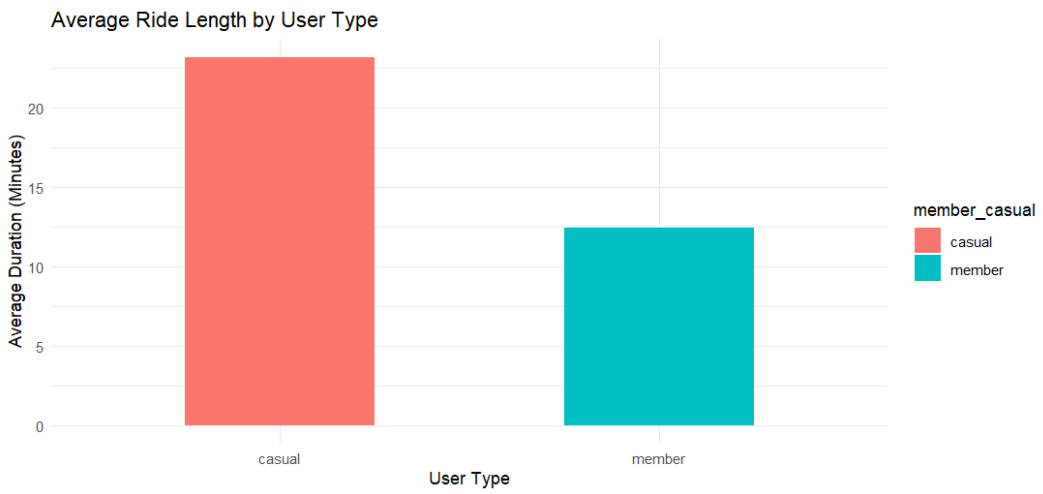
- **Electric Bike Preference:** There is a system-wide preference for electric bikes over classic, docked bikes, indicating that riders value speed and convenience.



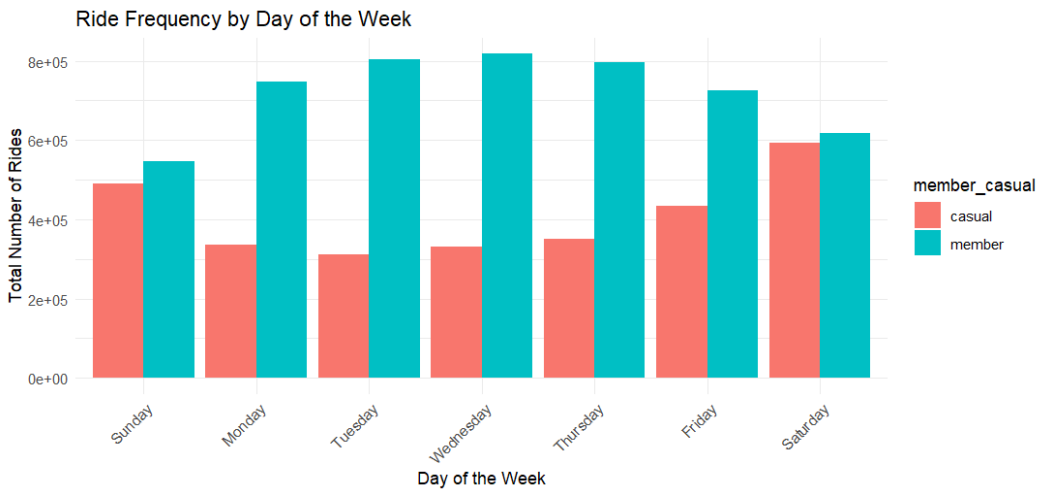
Behavioral Differences: Casual Riders vs. Annual Members

The data reveals two highly distinct user profiles. Annual members predominantly use Cyclistic as a practical utility for daily commuting, while casual riders use the service primarily for leisure and recreation.

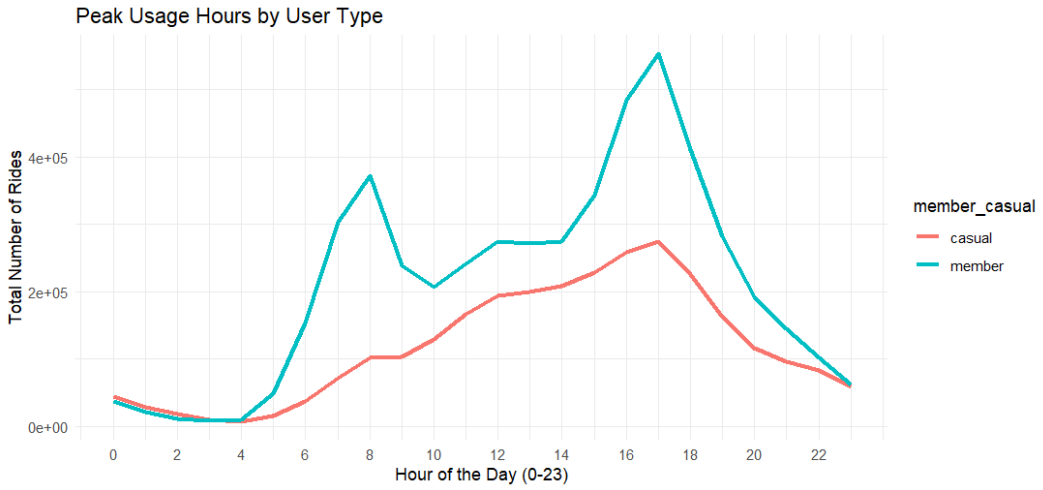
1. Trip Duration and Purpose Casual riders spend on average significantly more time on the bikes per trip compared to annual members. Members take short, efficient trips (likely point A to point B), whereas casual riders take long, extended rides (likely sightseeing or leisurely riding around the city).



2. Days of Operation Member usage is remarkably consistent from Monday through Friday, with a slight dip on the weekends. Conversely, casual rider usage is lowest during the workweek and spikes dramatically on Saturdays and Sundays.



3. Time of Day Member usage shows two sharp spikes at 8:00 AM and 5:00 PM, perfectly mirroring traditional corporate commuting hours. Casual ridership steadily builds throughout the morning and peaks mid-afternoon, completely lacking the morning rush-hour spike.



Phase 6: Act

Data-Driven Recommendations

Based on the behavioral differences identified between casual riders and annual members, the following actionable recommendations are presented to Lily Moreno and the Cyclistic Executive Team. The primary objective is to convert our highly active, leisure-focused casual riders into profitable, long-term annual members.

Target Audience Suggestions

Our data clearly indicates that the most promising segment for conversion is **local weekend riders and warm-weather enthusiasts**. Instead of trying to force casual riders to adopt the commuting habits of current members, we should tailor our messaging to resonate with their existing behavior: long, leisurely rides on weekends, particularly during the summer months.

Marketing Strategies & Membership Conversion Tactics

To bridge the gap between casual usage and annual commitment, I recommend the following three-pillar marketing strategy:

1. The "Summer Season" Push

- **The Data:** Casual ridership spikes dramatically from June to August and plummets in the winter.
- **The Tactic:** Launch an aggressive digital marketing campaign in late spring (May) offering a limited-time "Summer Kickoff" discount on the annual membership. The messaging should emphasize that purchasing an annual membership pays for itself after just a few long weekend rides during the summer months.

2. Duration-Based Value Marketing

- **The Data:** Casual riders take significantly longer trips than annual members.
- **The Tactic:** Create campaigns that highlight the financial benefits of the annual membership for long rides. Messaging such as, *"Stop watching the clock. Ride all afternoon for less with an Annual Membership,"* directly addresses the specific way casual riders already use the bikes, positioning the membership as a cost-saving tool for their weekend adventures.

3. "Weekend Warrior" Trial or Tier

- **The Data:** Casual riders primarily use the system on Saturdays and Sundays.
- **The Tactic:** Introduce a targeted promotion or a specialized "Weekend Only" annual pass. If a new tier is not feasible, Cyclistic could offer a one-month trial of the full annual membership marketed specifically at weekend riders to build habit and familiarity, increasing the likelihood of a full annual renewal.

4. The "Winter is On Us" Acquisition Campaign

- **The Data:** Casual riders avoid winter riding, creating a barrier to entry for annual commitment.
- **The Action:** Offer a limited-time "Introductory Extension" for *new* sign-ups in February: **"Pay for 1 year, get 15 months."**
- **Risk Mitigation:** To prevent friction with existing members, structure this as a **"New Member Welcome Offer."** Simultaneously, offer a **"Referral Bonus"** to existing members (e.g., a free month for every friend they convert), incentivizing them to bring casual riders into the fold rather than feeling alienated by the lower price.

Business Improvements Supported by Findings

Beyond external marketing, Cyclistic can leverage its internal systems and app to drive conversions:

- **In-App Savings Notifications:** Implement a feature in the Cyclistic app that tracks a casual rider's spending. After a casual rider completes a trip, send a push notification showing how much money they *would have saved* on that specific long ride if they were an annual member.
- **Targeted Email Workflows:** Identify casual riders who frequently rent electric bikes (our most popular bike type) on consecutive weekends. Trigger personalized email workflows offering them a customized discount code for an annual membership, leveraging their established riding habits.
- **Location-Based Advertising:** Since casual riders likely frequent parks, waterfronts, and recreational areas on weekends, Cyclistic should increase physical and digital ad placements near these high-traffic weekend start and end stations.
- Heavy-up digital advertising spend on **Friday afternoons and Saturday mornings**.